

DBE & DSD · PROJECT REVIEW 1 · 25CS1302E

# Enterprise Knowledge Intelligence Platform

Hybrid Relational, Document, and Vector Data Architecture

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# The Knowledge Fragmentation Problem

## Wasted Time

Employees spend significant time searching for information that should be readily accessible.

## Poor Search Quality

Traditional keyword-based search returns irrelevant results, failing to understand intent or meaning.

## Duplication & Cost

Knowledge duplication increases storage and maintenance costs across the organisation.

## Security Gaps

Lack of centralised management leads to version control issues and unauthorised access risks.

**Affected sectors:** IT Companies · Educational Institutions · Government Organisations · Healthcare Organisations · Research Institutes · Large Enterprises

# Why This Project Matters



## Beneficiary Groups

Employees, Managers, Researchers, Students, Organisations, Knowledge Admins

## Impact Areas

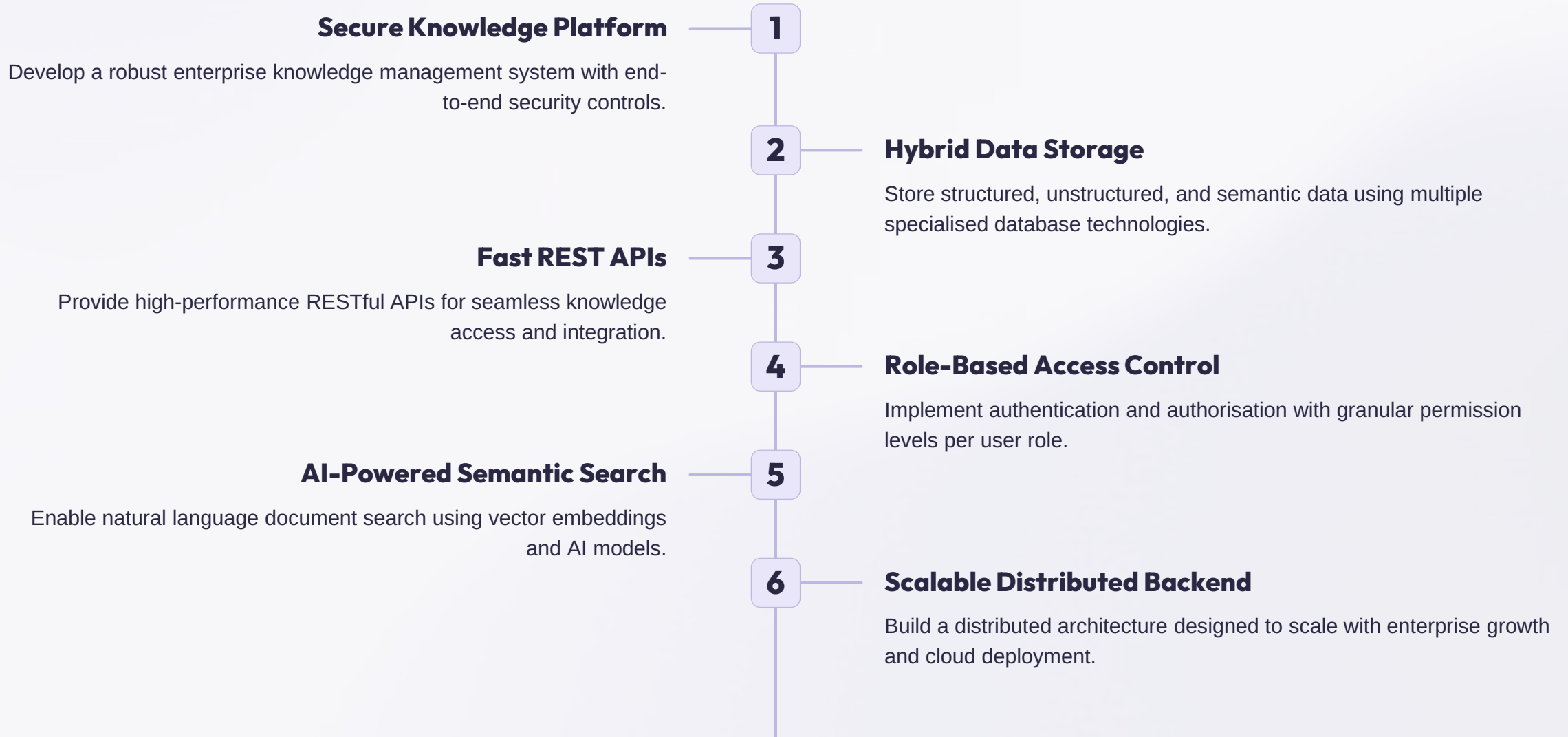
Productivity, Decision-making, Digital Transformation, Knowledge Sharing, Time Savings

Improve collaboration and information sharing across teams

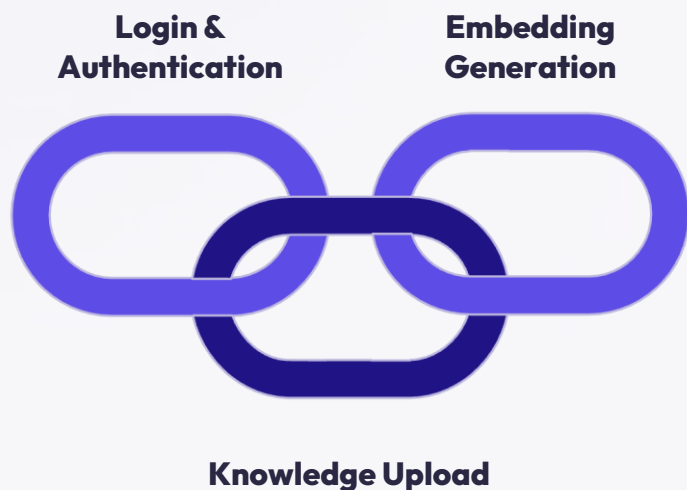
Centralise organisational knowledge in one secure platform

Replace keyword search with intelligent semantic understanding

# Project Objectives



# Proposed Solution & Methodology



## Main Modules

- User & Role Management
- Knowledge Repository
- Document & Semantic Search
- REST APIs & Authentication
- Deployment & Testing

## Database Stack

### PostgreSQL

Users, roles, permissions, and metadata

### MongoDB

Documents and knowledge articles

### Qdrant / Pinecone

Vector embeddings for semantic search

# Feasibility Analysis

## Hardware Requirements

### Public Dataset

#### Stack Exchange Data Dump

- Technical questions and answers
- Tags and metadata
- Used for semantic search and document retrieval

### Laptop / Desktop

Minimum Intel Core i5 / Ryzen 5,  
8 GB RAM, 20 GB free storage

### Operating System

Windows 10/11, Ubuntu Linux  
or macOS

### Internet Connection

Required for APIs & cloud services

## Software Stack

### Languages

Java · Python · JavaScript  
· SQL

### Frameworks

Spring Boot · React.js

### Databases

PostgreSQL · MongoDB ·  
Qdrant

### Tools

Docker · Git · Postman ·  
Maven · Visual Studio Code

### Security & AI

Spring Security · JWT · OpenAI / Sentence Transformers

# Novelty of the Project

## ✗ Existing Systems

Keyword search only —  
no semantic  
understanding

Single-database storage  
— limited flexibility

Poor scalability and  
version control

Minimal or no role-based  
access control

## ✓ Our Proposed System

### Hybrid Database Architecture

PostgreSQL + MongoDB +  
Vector DB working in  
concert

### AI Semantic Search

Search by meaning: *"How  
do I deploy the backend?"*

### Role-Based Security

Granular permissions for  
Admin, Manager, and  
Employee roles

### Scalable & Distributed

Microservices-ready,  
cloud-deployable, Docker-  
containerised

KEY INNOVATION

# Three Databases, One Intelligent Platform

The integration of **relational**, **document**, and **vector databases** in a single enterprise platform enables efficient data management, intelligent semantic search, enhanced security, and scalable architecture — making it significantly more powerful than conventional knowledge management systems.



## Relational Layer

PostgreSQL manages users, roles, permissions, and structured metadata with full ACID compliance.



## Document Layer

MongoDB stores flexible, schema-less documents and knowledge articles at scale.



## Vector Layer

Qdrant enables AI-powered semantic search by matching meaning, not just keywords.



# Architecture Overview

## System Architecture



## Design Principles

### Separation of Concerns

Each database handles the data type it is optimised for, ensuring performance and clarity.

### Security by Design

JWT-based authentication and role-based authorisation are embedded at every layer.

### Cloud-Ready

Docker containerisation and microservices patterns enable seamless cloud deployment.

# Summary & Next Steps

This project presents a novel, hybrid architecture that addresses real-world knowledge management challenges through modern database engineering and AI-powered search.

01

## Finalise Architecture Design

Confirm database schemas, API contracts, and embedding model selection.

02

## Set Up Development Environment

Configure Docker containers for PostgreSQL, MongoDB, and Qdrant.

03

## Implement Core Modules

Build authentication, document upload, and embedding generation pipelines.

04

## Integrate & Test

Connect all layers, validate semantic search, and conduct security testing.